Soft Suave
Technologies



Enhancing Cybersecurity with AI-Driven Intelligent Automation

About the Client

The client is a cybersecurity-focused enterprise aiming to streamline its threat detection, incident response, and compliance operations using advanced AI and LLM technologies. With a vision to replace static security workflows with autonomous, intelligent systems, the client sought to build a secure, scalable, and interactive AI-driven platform for enterprise cybersecurity.

Solution Overview

We developed a comprehensive AI-powered cybersecurity platform that leverages large language models (LLMs), real-time expert agents, graph-based intelligence, and dual RAG (Retrieval-Augmented Generation) pipelines to deliver dynamic, accurate, and secure cyber operations—entirely through a chat-first interface.

Key Challenges

- Fragmented and siloed cybersecurity data
- Inefficient manual incident response
- Limited contextual insight from documents
- Lack of automation in task execution

Client Requirement

- Chat-based access to cybersecurity data
- Real-time interaction with AI security experts
- Integration with structured playbooks and files
- Support for document retrieval and summary
- Visual representation of threat relationships
- Task automation using autonomous agents

Business Solutions

Multi-Expert Chat Interface

Implemented a real-time, modular chat system where users can interact with AI experts specialized in areas like SOC, IAM, GRC, and document processing, enabling streamlined communication and task execution.

Advanced Retrieval System

Integrated dual RAG pipelines—DuckDB for lightweight queries and Milvus with MinIO for large-scale semantic search—enhancing both performance and relevance of information retrieval.

Autonomous Agentic AI

Designed intelligent agents that break down user queries into subtasks, coordinate between domain-specific experts, and autonomously deliver high-quality, context-rich outputs.

Explore Section & Tool Integration

Built a centralized repository for security documentation with app-specific views and direct tool integrations (e.g., vulnerability scanners, log analyzers), enabling fast access to critical intelligence.

Neo4j-Powered Knowledge Graph

Enabled visual traversal of incidents, experts, attack patterns, and documentation through an interactive playground, improving threat comprehension and strategic analysis.

Use cases

Streamlining Incident Response in a Ransomware Attack

A ransomware attack threatened critical systems, prompting an urgent response from the cybersecurity team for containment and recovery.

Solution:
The team queried the AI platform, which instantly provided a tailored, step-by-step ransomware response playbook from the cybersecurity knowledge base.

Centralized Documentation and Knowledge Sharing Across Teams

A growing cybersecurity team faced inefficiencies due to scattered and outdated documentation.

Solution:
The Explore Section centralized all cybersecurity documentation, making it searchable via DuckDB for quick access. It also integrated external tools like Nessus for real-time querying, improving document retrieval.

Proactive Threat Intelligence and Vulnerability Management

A global company struggled with staying ahead of emerging threats and managing vulnerabilities across its network.

Solution:
The platform's advanced RAG system and integrations (Milvus, Splunk) allowed real-time analysis of security bulletins and vulnerability databases. It correlated alerts with internal and external threat intelligence, using Neo4j to visualize connections between vulnerabilities and attack patterns.

Results & Impact

75% reduction in manual triage and data retrieval time

2x faster incident response through automated expert task handling

80% improvement in report accuracy using document-backed LLM outputs

Conclusion

By integrating cutting-edge LLM interaction, vector retrieval, autonomous agents, and graph-based visualizations into one unified platform, the client transformed how enterprise cybersecurity operations are executed. Soft Suave played a pivotal role in architecting and delivering this intelligent, scalable solution—empowering security teams with faster insights, improved automation, and a truly modern user experience.